ANALYTICAL METHODS

1 **DESCRIPTION**

1.1 Physically examine a representative sample of the material for Appearance and other characteristics: it is A white or yellowish, crystalline powder, slightly soluble in water, in acetone, in alcohol and in methylene chloride..

2 **IDENTIFICATION**

2.1 Melting Point

Determine the melting point using the capillary method: **It melts at between 159 - 163°C**

2.2 Light Absorption

2.2.1 0.1M hydrochloric acid (Preparation)
Measure 8.5mL of concentrated hydrochloric acid into 1000mL volumetric flask containing about 600ml of distilled water and make to the mark with distilled water.

2.2.2 Sample preparation

2.2.3 Accurately weigh and transfer about 40mg of sample into a 100ml volumetric flask.

2.2.4 Add 60ml of 0.1M hydrochloric acid, mechanically shake/sonicate till dissolution is complete.

2.2.5 Dilute to the mark with 0.1M hydrochloric acid and mix.

2.2.6 Pipette 5.0ml of this solution into another 100ml volumetric flask, dilute to the mark with 0.1M hydrochloric acid and mix.

2.2.7 Using a suitable spectrophotometer and 1-cm cells, measure the absorbances between the range of 230nm and 350nm:
a] Solution shows absorption maximum, at 277nm and a minimum at 240nm.
b]The specific Absorbance = $\frac{\text{Maximum Absorbance X 100 X 100}}{\text{Sample Weight (g) X 5 X 100}}$

Specific Absorbance is between 365 to 395

2.3 Aromatic Amines

2.3.1 5%w/v 2-Napththol Solution (Preparation)
Weigh and transfer about 5g of 2-Naphthol into a 100ml volumetric flask.
Add 40ml of 2M sodium hydroxide solution, mechanically shake/sonicate till dissolution is complete
Dilute to the mark with distilled water and mix

2.3.2 Procedure
2.3.3 Accurately weight and transfer about 10mg of sample into 50ml beaker.
2.3.4 Add 10mg of zinc powder, 1ml of distilled water and 0.25ml of 2M hydrochloric acid and swirl to moisten the powder mixture.
2.3.5 Transfer onto a steam-bath and heat for 5 minutes.
2.3.6 Cool to room temperature.
2.3.7 Add 0.2ml 10%w/v solution of sodium nitrite and mix.
2.3.8 Allow to stand for 2 minutes add 1ml of 5%w/v 2-naphthol solution and mix: **An intense orange or red colour and, usually, a precipitate of the same colour are produced.**

3 Appearance Of Solution

3.1 Accurately weigh and transfer about 1g of the sample into a 20ml volumetric flask.
3.2 Add 15ml of 1M hydrochloric acid, mechanically shake/sonicate till dissolution is complete.
3.3 Dilute to the mark with 1M hydrochloric acid and mix: **The solution is not more opalescent than reference solution II and not more intensely coloured than reference solution.**

4 Loss On Drying

4.1 Weigh labelled petri dish (W1) that has previously been dried at 105°C and cooled
4.2 Accurately weigh about 1g onto a petri-dish, ensuring they are evenly spread, then

Determine their weight together (W2).

4.3 Transfer into an Oven operating at 105°C.

4.4 Dry for 3 hours.

4.5 Transfer into a desiccators to cool room temperature.

4.6 Weigh the petri-dish with its content (W3).

Calculation:

$$\% \text{ Moisture} = \frac{(W2 - W3) \times 100}{(W2 - W1)}$$

Where, W1 = weight of empty petri-dish, g.

W2 = weight of sample and petri-dish, g.

W3 = weight of petri-dish and the dried sample, g.

5 SULPHATED ASH

5.1 Weigh two crucibles (W1) that has previously been ignited and cooled.

5.2 Accurately weigh about 1g onto each crucible, ensuring they are evenly spread, then determine their weight together (W2).

5.3 Heat, gently at first on a Bunsen flame, until the sample is thoroughly charred.

5.4 Cool, then moisten the residue with 1mL of sulphuric acid, heat gently until white fumes are no longer evolved, and ignite at 800 ± 25°C until the carbon is consumed (about 2 hours).

5.5 Cool in a desiccator, weigh, and calculate the % Residue on ignition, if the % Residue on ignition exceed 0.1%, again moisten the residue with 1ml of sulphuric acid, heat and ignite as before.

5.6 Continue until constant weight is attained (until two successive weighing does not differ by more than 0.5mg).

5.7 Weigh the crucibles with its content (W3).

5.8 Calculation:

$$\% \text{ Sulphated Ash} = \frac{(W3 - W1) \times 100}{(W2 - W1)}$$

Where, W1 = weight of empty petri-dish, g.

W2 = weight of sample and petri-dish, g.
W3 = weight of petri-dish and the dried sample, g.

6 **HEAVY METALS**

6.1 Lead stock solution(100ppm)

6.1.1 Accurately weigh and transfer about 160mg of lead into 250ml beaker.

6.1.2 Add 100ml of de-ionized water, 1ml of nitric acid and mix till dissolution is complete.

6.1.3 Rinse the 250ml beaker with several quantities of de-ionized water into the 1-litre volumetric flask.

6.1.4 Dilute with distilled water to 1000ml and mix. Prepare and store this solution in glass containers free from soluble lead salts.

6.2 Lead stock solution (10ppm)

6.2.1 Pipette 2.0ml of the 100ppm lead standard solution into a 20ml volumetric, dilute to the mark with methanol and mix.

6.3 Buffer Solution pH3.5

6.3.1 In a 250ml beaker, dissolve 25.0g of ammonium acetate in 25ml of water and add 38.0ml of 7M hydrochloric acid.

6.3.2 Adjust the pH if necessary with 2M hydrochloric acid or 6M ammonium hydroxide and dilute to 100.0ml with water.

6.4 Thioacetamide SOLUTION(4%)

6.4.1 Dissolve 4g of thioacetamide in a 100ml volumetric flask, make up with distilled water

6.4.2 Thioacetamide Reagent

6.4.3 To 2ml of thioacetamide solution in a 100ml volumetric flask, add 7.5ml of water, 37.5ml of 1 M sodium hydroxide and 50ml of glycerol (85%). Heat in a water-bath for 20 seconds. Prepare immediately before use.

6.5 Sample Preparation

6.5.1 Accurately weigh and transfer about 1g of sample into a 20ml volumetric flask.

6.5.2 Add 15ml of distilled water, mechanically shake/sonicate till dissolution is complete.

6.5.3 Dilute to the mark with distilled water and mix.

6.5.4 To 12ml of the solution thus obtained (in a test tube), add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

6.6 Standard Preparation

6.6.1 Transfer 10 ml of 1ppm lead standard solution and 2 ml of the sample solution into a suitable test tube.

6.6.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

6.7 Blank Preparation

6.7.1 Prepare a blank using a mixture of 10ml of water and 2ml of sample solution.

6.7.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

6.8 Observation

6.8.1 Compared to the blank, after 2 minutes any brown colour in the solution obtained with the sample is not more intense than in the standard solution

7 Assay

7.1 **Method A**

7.1.1 **Reagents**

7.1.2 0.1M perchloric acid; prepared

7.1.2 0.1% crystal violet or naphthobenzein in glacial acetic acid

7.1.3 Accurately weigh and transfer about 0.1g to 100ml volumetric flask, dilute to the mark with glacial acetic acid and mix

7.1.4 Sample preparation

7.1.5 Accurately weigh about 150mg each into two separate, dry 250ml conical flasks.

7.1.6 Add 50 ml of glacial acetic acid to each flask and heat on a hot plate (with swirling from time to time) till dissolution is complete.

7.1.7 Add two drops of 0.1% of crystal violet in glacial acetic acid and titrate with 0.1M perchloric acid in glacial acetic acid to an emerald green end-point.

7.1.8 Calculate the content as follows:

$$\%Assay = \frac{\text{Titre X 17.12mg X Standard factor of } HClO_4 \text{ X 100}}{\text{Sample weight (mg)}}$$

7.2 **Method B**

7.2.1 **Reagent**

1M hydrochloric acid

Measure 85mL of concentrated acid hydrochloric acid into 1000mL volumetric flask containing 600ml distilled water, and make to the mark with distilled water

7.2.3 Sample Preparation

7.2.4 Accurately weigh and transfer about 70mg of metronidazole sample into a 100ml volumetric flask.

7.2.5 Add 60ml of 1M hydrochloric acid mechanically/sonicate till dissolution is complete.

7.2.6 Dilute with 1M hydrochloric to volume and mix well.

7.2.7 Pipette 2.0ml of this solution into another 100ml volumetric flask, dilute to the mark with 0.1M hydrochloric acid and mix.

7.2.8 Procedure

7.2.9 Concomitantly determine the absorbances of assay preparations in 1-cm cells at Wavelength of maximum absorbency at about 276nm, using 1M hydrochloric acid as blank And A1% as 375

8 Microbiological Tests

8.1 Limulus Amebocyte Lysate LAL Test

8.1.1 Open the seals of the test tubes of lyophilized LAL Reagent

8.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

8.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

8.1.4 Incubate the reaction mixture immediately at 37°c for 60minutes. Avoiding vibration during incubation.

8.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

8.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

8.1.7 Read reaction tubes one by one.

8.1.8 Invert the tube in one smooth motion.

8.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by the absence of solid clot after inversion.

8.2 Testing Methodology of Bacterial Endotoxin (LAL)

8.2.1 Each specimen shall be tested in duplicate

8.2.2 NMC: Take 100µl(0.1ml) LRW

8.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

8.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

8.2.5 Add the LAL reagents as mentioned below in the table 1

8.2.6 Mix gently and incubate the tubes in heating block at 37^0C ± 1 for 60min.

8.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 180^0C and note the observations.

8.2.8 If a firm gel has formed, record the results as positive.

8.2.9 A result is negative, if an intact gel is not formed.

8.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

8.3 Pyrogen

8.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed.

8.3.2 Withhold all food from the rabbits used during the period of the test

8.3.3 Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring.

8.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit.

8.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm.

8.3.6 If the initial temperature is less than 38.8^{0}c or more than 39.8^{o}cexclude the animal from the test.

8.3.7 Ensure all the needles and syringe to be used are new

8.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 1^{0}c.

8.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection.

8.3.10 Any rabbits showing more than 0.2^{0}c temperature difference between two successive readings in the last three readings can be excluded from the test.

8.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes.

8.3.12 Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution.

8.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected.

8.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

8.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution

8.4 MLT

8.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

8.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

8.4.3 Testing to be performed under LAF of specified area

8.4.4 Arrange all the required material on the LAF before starting the test.

8.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

8.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45°C) in two petri plates for TAMC

8.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45°C) in two petri- plates for TYMC

8.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate

8.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45°c and allow for solidification.

8.4.10 Incubate the plates at 30°C to 35°C for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 20°C to 25°C for 5 days in an inverted position.